THE PROBLEM

Robots are about to work for hire. Their track record will belong to whoever runs the platform.

A failed job can be deleted, a score edited, an incident hidden. Customers, robot owners and the people who sell AI agents to robots have no shared record none of them controls, and stale bookings and offers pile up in someone's database.

WHAT WE BUILT

Semmal: robots that sign their own record on Arkiv

1

The robot owns its record

Quotes, the agents it hires, every job and every incident are Arkiv entities owned by the robot's own wallet.

2

Scores pick who gets work

Each job's score is an indexed column; it ranks robots for customers and agents for robots, and anyone can recompute it.

3

Expiry is the coordination

Bookings, quotes and job claims expire at their own deadline: no stale offers, no clean-up.

DELIVERABLE 1 · THE ROBOT OWNS ITS RECORD

Nobody, not even its owner, can edit or withhold it

Key made on the robot
never leaves the robot PC; may write only its 4 evidence schemas, under a daily fee cap

→

Arkiv entity
$owner = $creator = robot wallet, and the body names the same wallet

→

Readers enforce it
a record counts only if that wallet is on the robot's registered profile

Live: **7** episodes and **21** incidents signed by robot wallet `0x103E…29BC`. There is no platform key anywhere.

DELIVERABLE 2 · SCORES FROM ARKIV CHOOSE WHO GETS WORK

One query, and anyone gets the same answer

```
$ pnpm arkiv:query
task-6f5992c2  COMPLETED  task_score 100
owner 0x103E…29BC (robot wallet)
  Stops completed      40/40  4 of 4 stops
  Valid 0G decisions   20/20  4 of 4 passed
  Findings dealt with  20/20  1 of 1
  On time              10/10  before deadline
  agent agt-house-2b5c20 · 4 jobs · 100 each
```

Indexed `task_score`

The robot scores each job itself, and every agent it paid. Robot and agent reputations are the mean of those signed scores.

They rank robots when a customer books, and agents when a robot hires. A safe stop never counts against a robot.

DELIVERABLE 3 · EXPIRY DOES THE COORDINATION

Stale offers clean themselves up

deadline

Bookings, quotes, job claims

expire exactly when they stop mattering.

90 d

Robot and agent registrations

unmaintained or badly scored ones simply lapse.

120 d

Evidence and scores

recent work is scored; full bundles stay on 0G Storage.

Every booking shows **"What went to Arkiv"**: the entity read back, its columns, body, owner, expiry and the score maths.

WHY ARKIV

Swap Arkiv for Postgres and users lose a robot record they don't have to trust us for.

arkiv/*.md · schema, friction, reach

pnpm arkiv:query

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